

REVIEW OPEN ACCESS

Function and Spatial Organization of Tumor-Invasive Human $\gamma\delta$ T Cells—What Do We Know?

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Correspondence: Daniela Wesch (daniela.wesch@uksh.de)**Received:** 2 July 2024 | **Revised:** 15 November 2024 | **Accepted:** 18 November 2024**Funding:** This work was supported by the Forschungsgruppe FOR2799 “Receiving and Translating Signals via the $\gamma\delta$ TCR” from the Deutsche Forschungsgemeinschaft (DFG, WE 3559/6-2 to DW and WI 5021/2-2 to KWH).**Keywords:** cancer immunotherapy | checkpoint inhibitors | tumor | tumor-infiltrating gammadelta T cells | tumor microenvironment

ABSTRACT

Human gammadelta ($\gamma\delta$) T cells not only infiltrate or reside in healthy tissues but also enter solid cancers. A large body of evidence suggests that $\gamma\delta$ T cells can exert potent anti-tumor effects, although conflicting or unfavorable effects have been reported in some cancer entities. Infiltration patterns are key to understanding the complexity of the tumor microenvironment (TME) and its interplay with $\gamma\delta$ T cells. The limited data available describe different $\gamma\delta$ T cell subsets that are located in different areas around and within tumors. Tumor-infiltrating $\gamma\delta$ lymphocytes ($\gamma\delta$ TIL) exert cytotoxicity, for example, via the CD95- or TRAIL-axis, produce high amounts of granzymes, and after their activation, tumor necrosis factor (TNF)- α or IFN- γ and express immune checkpoint receptors. Under certain conditions, $\gamma\delta$ T cell subsets can express low amounts of IL-17 and seem to contribute to immune regulation/suppression. A polarization of $\gamma\delta$ T cells can be influenced by the TME. Inflammatory cytokines, growth factors, or tumor promoters can suppress $\gamma\delta$ T cell functionality or even push them toward tumor promotion. To avoid this and to exploit the unique features of $\gamma\delta$ T cell-mediated anti-cancer and immune-orchestrating capabilities in future immune therapy approaches, a growing body of preclinical but also clinical studies can be observed.

1 | Introduction: Gammadelta ($\gamma\delta$) T Cells in Solid Cancers

Human $\gamma\delta$ T lymphocytes are a heterogenous subpopulation of lymphocytes that have the ability to infiltrate solid tumors. They seem to play a non-redundant role in tissue homeostasis and carcinogenesis [1]. Although $\gamma\delta$ T cells usually account for only 2%–10% of the total peripheral blood T cell population, their proportion is altered in diseases and is higher in some tissues. Depending on the expression of their V δ chain, $\gamma\delta$ T cells are clas-

sified in at least three major subsets, in V δ 1, V δ 2, and V δ 3 T cells. Although V δ 1 and V δ 3 T cells are found mainly in the mucosa, the skin, and the latter also in the liver, V δ 2 T cells dominate the blood. Interestingly, all of these $\gamma\delta$ T cells subsets infiltrate solid tumors [2]. Mounting clinical interest in exploiting these unconventional T cells for cancer immunotherapies justifies the need for a better understanding of the roles and functions of $\gamma\delta$ T cells [3]. This review is focused on human intra-tumoral $\gamma\delta$ T cells in solid tumors and presents an update of their localization and functional capacity in the tumor microenvironment (TME).

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2 | Localization of Tumor-infiltrating $\gamma\delta$ T lymphocytes ($\gamma\delta$ TILs)

$\gamma\delta$ T cells reside and infiltrate in epithelial tissues, where many solid tumors develop. Tumor-infiltrating $\gamma\delta$ T lymphocytes ($\gamma\delta$ TILs) were identified in numerous solid cancers, for example, melanoma, colorectal, liver, breast, pancreatic, and ovarian cancers [4–10]. These cells are often associated with high cytotoxicity and anti-tumor functions [11, 12], but they also appear to be able to promote tumor formation under certain conditions [13–15]. Thus, a better understanding of the TME, its composition, and modulatory effects is required to decode the underlying mechanisms to better understand contradictory findings.

Along these lines, diverse infiltration patterns could be identified in different entities. For example, $\gamma\delta$ TILs were distributed always in the periphery of the infiltrate in a small study of melanoma patients and were composed of both the V δ 2 and V δ 1 T cell subsets [5]. Higher proportions of $\gamma\delta$ TILs were linked to prolonged patient survival [11]. In cervical cancer, $\gamma\delta$ TILs correlated positively with clinical outcome and were associated with a lower presence of suppressive and higher cancer-rejecting immune cells in the TME [12]. But in colorectal cancer (CRC) and hepatocellular carcinoma (HCC), $\gamma\delta$ T cell abundances seem to be higher in the tumor/stroma border than in the intra-tumoral tissue [6, 16]. In addition, lower $\gamma\delta$ TILs counts in the tumor stroma correlated with more frequent recurrence in HCC patients [16]. The involvement of $\gamma\delta$ and other innate-like T cells in liver diseases, some of which may be precursors to cancer, has recently been reviewed by us, summarizing the pro- and anti-inflammatory roles of certain $\gamma\delta$ T cell subsets [10]. In contrast, Th17-like $\gamma\delta$ TILs in CRC expressing dominantly the V δ 1 T cell receptor (TCR) have been described to recruit and induce proliferation and survival of polymorphonuclear myeloid-derived suppressor cells (PMN-MDSC) and contribute thus to cancer progression [13]. Joint immune suppression of $\gamma\delta$ T cells and myeloid-derived suppressor cells (MDSCs), mesenchymal stem cells (MSC), and tumor-associated macrophages (TAM) is described [17–19]. $\gamma\delta$ T cell immune response can be improved by the inhibition of cyclooxygenase 2 (COX-2) activity, which suppresses MDSC-, MSC-, and TAM-associated suppressive factors such as arginase I expression or IL-10 and TGF- β , prostaglandin (PG) release, or ROS production [18–20].

In contrast to these findings in CRC, $\gamma\delta$ T cells are localized in both the adjacent tissue and the malignant ductal epithelium in pancreatic ductal adenocarcinoma (PDAC) [8, 21, 22]. Interestingly, $\gamma\delta$ T cells seem to co-localize with Tregs, suppressive fibroblasts, and CD163⁺ M2-macrophages as well as with CD68⁺ macrophages in the ductal/tumoral epithelium. Carcinoma cells in PDAC showed a significant enhanced proliferation activity indicated by a high percentage of Ki67⁺ cells compared to ductal epithelial cells in chronic pancreatitis patients [22]. Again, controversial pro- as well as anti-tumoral associations of $\gamma\delta$ T cell infiltration were reported in breast cancer [4, 14, 23]. Recently, emerging multiplexed tissue imaging techniques allow the detection of more than 150 protein markers on a single tissue slide with subcellular resolution, including $\gamma\delta$ T cell subsets [24], which will enable us in the future to gain new insights into the complex interplay among $\gamma\delta$ T cell phenotype, localization, and functionality with and within the TME.

3 | Functional Properties of Tumor-infiltrating $\gamma\delta$ T lymphocytes ($\gamma\delta$ TILs)

Intra-tumoral $\gamma\delta$ T cells seem to be in a pre-activated stage because they intrinsically produce higher amounts of granzymes compared to the $\gamma\delta$ T cells isolated from the blood of the same donors [7]. Additionally, the number of effector memory V δ 1 and V δ 2 TIL is increased in comparison to peripheral blood lymphocytes [7, 25–27]. In striking contrast to the intrinsic granzyme production by $\gamma\delta$ TIL, production of cytokines such as IFN- γ and tumor necrosis factor (TNF)- α or IL-17 is induced after TCR ligand stimulation of $\gamma\delta$ TIL and influenced by the surrounding cytokine milieu [6, 7, 13, 28]. The intestinal microbiome or an accumulation of certain pathogens after epithelial barrier damage in CRC can differentially influence $\gamma\delta$ TIL polarization [29]. Additionally, a desmoplastic stroma in pancreatic cancer contains many immunosuppressive cells, which also has an influence on polarization of $\gamma\delta$ TIL [22]. Future functional investigations may benefit from our novel assay, which enables the detection of $\gamma\delta$ T cell activation as early as 10 min after stimulation via detection of a conformational change of the lymphocyte function-associated antigen-1 [30].

The antigen recognition of V δ 1 and V δ 3 TILs, which are often enriched at the tumor site, is not well defined and is not limited to the TCR [2, 7, 31]. Expression of miscellaneous surface molecules on V δ 1 TIL plays a role in their immune cell regulation or tumoral activity. For example, several cancer patient-derived V δ 1 T cell lines directly bind via natural killer group 2 member (NKG2) D to stress-induced MHC class I-related chain (MIC) A and UL16-binding protein (ULBP)-3 expressed on tumor cells [32, 33]. Intra-tumoral ecto-5'-nucleotidase-(CD73) expressing V δ 1 TIL in breast cancer patients is suggested to suppress anti-tumor responses, whereas V δ 1 TIL in ovarian cancer patients does not express CD73 (Figure 1Aa,e) [15, 34]. Instead of CD73, which can induce exhaustion of cells, ovarian naive and central memory V δ 1 TIL highly express exhaustion markers and immune checkpoint markers such as T cell immunoreceptor with Ig and ITIM domains (TIGIT) and programmed cell death protein (PD-1) (Figure 1Ba) [34, 35]. After the activation of ex vivo isolated V δ 1 TIL, effector memory cells as well as TIGIT and PD-1 expression increased. In contrast, ex vivo isolated ovarian V δ 2 TIL only slightly expresses TIGIT and PD-1 (Figure 1Ba), which were transiently increased after V δ 2 TIL activation [35]. Both ovarian $\gamma\delta$ TIL subsets can lyse autologous tumor cells after stimulation by appropriate bispecific antibodies (bsAb), which enhance $\gamma\delta$ T cell cytotoxicity and overcome their immunosuppressive capacity [7, 35].

TIGIT is also expressed by cytotoxic V δ 1 TIL in CRC patients with putative anti-tumoral capacity (Figure 1Bb). It has recently been described that V δ 1 TIL in microsatellite-stable CRC displays a dysfunctional but not exhausted effector state, mediated via the TIGIT-NECTIN2 axis by immunosuppressive fibroblasts. Manipulation of such dysfunctional V δ 1 TIL with antagonistic TIGIT antibodies restored partially cytotoxicity in vitro, underlining that also these unconventional T cells can be direct targets of immune therapies [36]. Similar to this, cytotoxic PD1⁺ V δ 1 TIL contributes to the response to PD-1 and CTLA-4 blockade in patients with HLA-class I-negative MMR-d colon cancers, while autologous

Figure 1 A

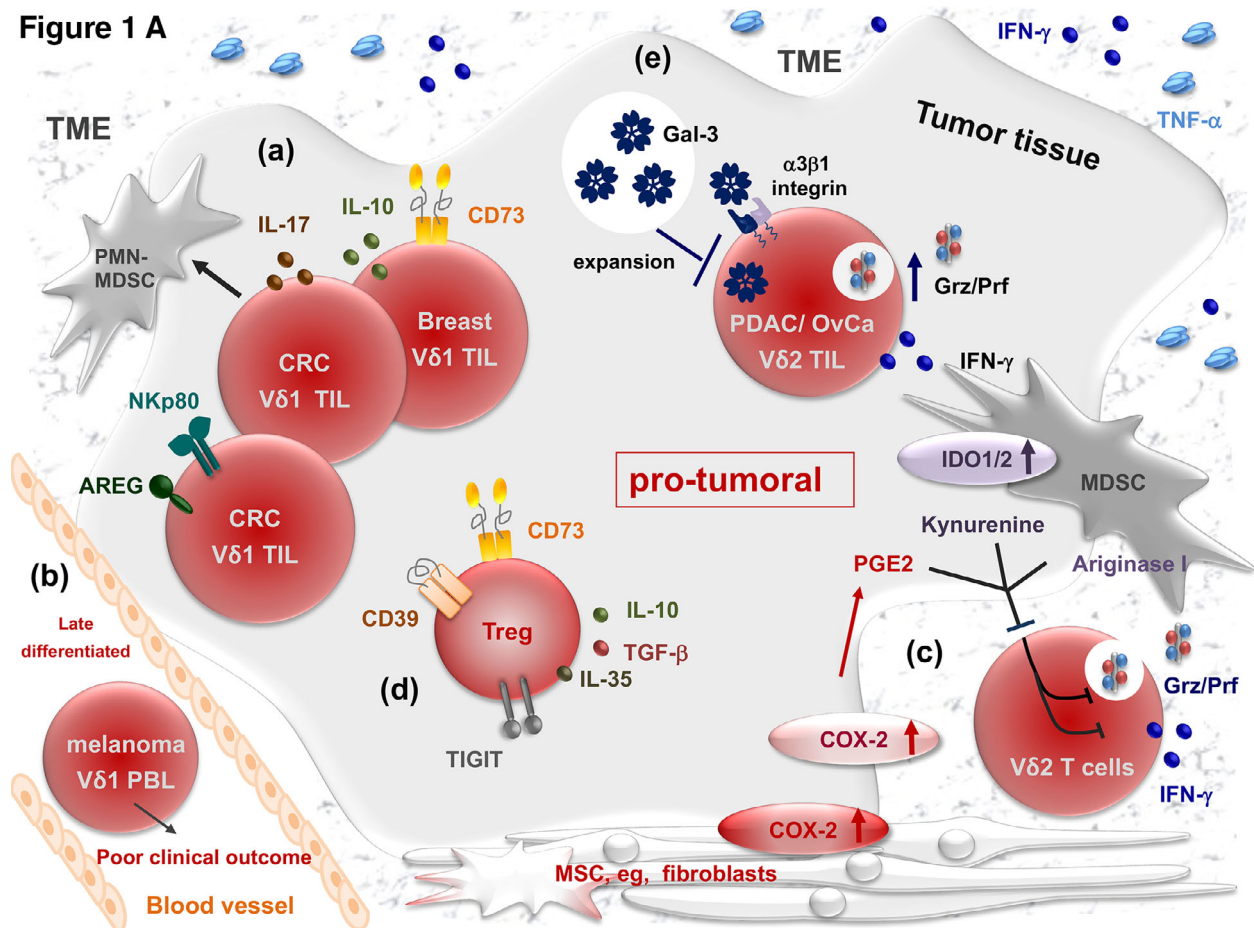


FIGURE 1 | $\gamma\delta$ TIL of different tumor entities and the influence of TME. (A) Pro-tumoral activity—(a) V δ 1 TIL with pro-tumoral capacity is described in breast and colorectal cancer (CRC) patients. CD73-expressing V δ 1 TIL of breast cancer patients releases IL-10, adenosine, and IL-8, whereas rare IL-17-producing V δ 1 TIL produces IL-8, TNF- α , and GMC-SF and influences polymorphonuclear myeloid-derived suppressor cells (PMN-MDSC) in CRC patients. NKp80-expressing V δ 1 TIL of CRC patients release suppressive amphiregulin (AREG). (b) High V δ 1 T cell frequencies in peripheral blood lymphocytes (PBLs) of melanoma patients are associated with poor clinical outcome and a late differentiated phenotype. (c) Tumor cells and immunosuppressive stromal cells of the TME such as MDSC or MSC overexpress enzymes like cyclooxygenase (COX) or indoleamine-2,3-dioxygenase (IDO), which mediate the enhanced release of prostaglandin (PG) E2 or kynurenine, respectively. The latter suppresses anti-tumor response of $\gamma\delta$ TIL, comparable to arginase I does. (d) Regulatory T cells (Treg) highly infiltrate the different tumor entities and exert their immunosuppressive effects on T cells by the indicated pathways. (e) On the basis of their enhanced expression of galectin-3-binding surface receptor $\alpha 3\beta 1$ integrin, V δ 2 TIL of pancreatic and ovarian cancer patients is inhibited in their proliferation after cross-talk with gal-3-producing tumor cells. (B) anti-tumoral activity—(a and b) V δ 2 TIL of breast, prostate, pancreatic, and ovarian cancer patients and cytotoxic V δ 1 TIL of CRC patients exert their anti-tumor response by the release of IFN- γ , TNF- α , TRAIL, granzymes, and perforin. (a) Although granzymes are permanently enhanced in V δ 1 and V δ 2 TIL of ovarian cancer patients, other cytokines are induced after the TIL activation. An enhanced expression of PD-1 and TIGIT on ex vivo isolated V δ 1 TIL, in contrast to V δ 2 TIL of ovarian cancer patients, is described and is up-regulated on both subsets after their stimulation. (c) TRAIL-R4 has an important anti-tumoral role on V δ 1 and V δ 2 T cell subsets by regulating COX-1/2. (d) V δ 1 TIL are not influenced in their proliferation by gal3-producing tumor cells due to their low expression of galectin-3-binding surface receptor $\alpha 3\beta 1$ integrin. MSC, mesenchymal stem cells.

CD8 T cells failed to respond in these patients with genomic inactivation of $\beta 2$ -microglobulin [37].

However, there is also literature describing opposing roles of V δ 1 TIL in CRC. For example, amphiregulin (AREG)-producing V δ 1 TIL with pro-tumoral function have recently been described in CRC patients (Figure 1Aa) [38]. AREG interacts with the Epidermal growth factor receptor (EGFR) to promote the growth and differentiation of normal epithelial cells, but an AREG overexpression in colon or ovarian cancer patients can induce tissue progression or chemotherapy resistance, respectively. The blockade of AREG production is suggested to induce greater

tumor clearance and infiltration as well as diminish evolving secondary chemotherapy resistance mechanisms [38, 39].

In the blood of melanoma patients with poor clinical outcome, an increased V δ 1 T cell compartment has been associated with poor patients' survival before the advent of immune checkpoint blockade (ICB) and under CTLA-4 inhibition (Figure 1Ab) [40–42]. In the latter case, we identified a differentiation signature in patients with high V δ 1 T cell frequencies that was dominated by late differentiated phenotypes [43]. Recent single-cell RNA sequencing of melanomas revealed enrichment and reprogramming of $\gamma\delta$ TILs in clusters in response to ICB and varied largely

Figure 1B

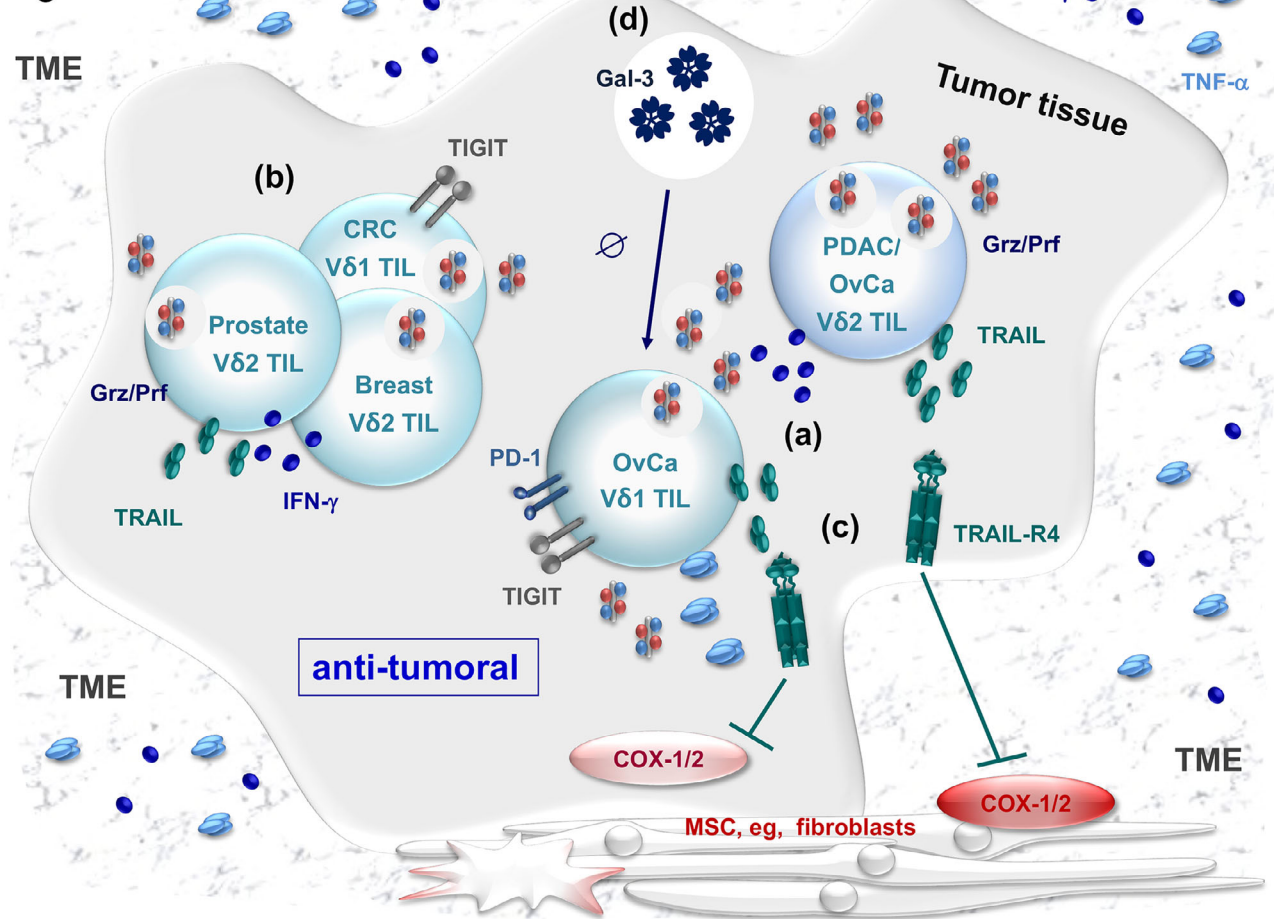


FIGURE 1 | (Continued)

between Vδ1 and Vδ2 T cells. Low expression of LAG3, HAVCR2, and PDCD1 is associated with improved patient survival [44].

Vδ2 TIL, which mainly co-expresses a Vγ9 chain, recognizes pyrophosphate intermediates of a dysregulated mevalonate pathway in tumor cells by a butyrophilin (BTN) molecule-dependent mechanism [45–47]. Pyrophosphate intermediates induce a conformational change of BTN3A1 (CD277), which is a widely expressed surface cell protein and highly expressed on tumor cells. The activation of BTN3A1 allows the binding of BTN2A1 to an invariant germline-encoded region of Vγ9, thereby activating Vγ9Vδ2 T cells [45–47]. An increased expression of the BTN2A1–BTN3A complex accompanied by an enhanced Vγ9Vδ2 TCR-mediated killing induced by an activation of AMP-activated protein kinase is described in a cell line model and in patient-derived tumor organoids [48].

Cytotoxic activity of intra-tumoral γδ T cells can also be mediated by NKG2D binding to MICA expressed by tumor cells in a TCR-independent or TCR-dependent pathway [49].

Besides granzymes and cytokines, γδ TIL exerts their cytotoxicity via the CD95/CD95L or the TNF-related apoptosis-inducing ligand (TRAIL)—TRAIL receptor pathway [2]. In addition, recent data revealed that FAS-L expression of γδ TIL can mediate the death and subsequent disappearance of γδ TIL after interaction

with the tumor-surrounding cells. This interaction can induce “cold” tumors that fail to respond to immunotherapy [50]. Several other studies demonstrated that Vδ2 TIL cytotoxicity can be mediated by TCR- or NKG2D-dependent release of soluble TRAIL, whereas a minor role of the TRAIL/TRAIL-R pathway is suggested for Vδ1 TIL [51–54]. Interestingly, TRAIL-R4 (decoy receptor 5, DcR5), which is unable to induce apoptosis due to the lack of a pro-apoptotic death domain, is described to have pro-tumoral functions. In contrast to the described pro-tumoral function of TRAIL-R4, knockdown of anti-apoptotic TRAIL-R4 in breast and pancreatic cells reduced the granzyme production by γδ T cells and rendered them more resistant to lysis of these tumor cells (Figure 1Bc). Our results suggest an important role of TRAIL-R4 in the anti-tumoral cytotoxic activity of γδ T cells [54].

Further, the functional properties of γδ TIL are presumably influenced by polymorphisms of regulatory genes, genetic differences among tumor cells, localization of tumors, and the TME.

4 | Influence of the Tumor Microenvironment

The TME influences the activation and polarization of γδ TIL, and thereby their pro- or anti-tumor response. An immunosuppressive TME can hinder γδ TIL to exert anti-tumor response and can promote γδ T cell polarization to an immunosuppressive pheno-

type. Immunosuppressive cells, including an enhanced number of regulatory T cells (Treg), tumor-associated macrophages (TAM), MDSC, or mesenchymal cells (MSC) such as cancer-associated fibroblasts, can prevent the activation and cytotoxicity of $\gamma\delta$ TIL [2, 25].

Pro-inflammatory cytokines, such as TNF- α , growth factors, tumor promoters, and TRAIL-R4 knockdown, enhance COX-2 expression in different tumor entities in cancer-associated fibroblasts and other MSC cells [18, 20, 54]. The latter are involved in the development of tumor-promoting cancer stroma [55]. An enhanced COX-2 enzyme expression results in an increased release of its metabolites such as PG E2, which suppresses anti-tumor response of $\gamma\delta$ T cells and recruits Treg (Figure 1Ac,d). Treg can suppress $\gamma\delta$ T cells in different ways, as follows: (i) by the release of soluble inhibitory cytokines, such as IL-10, TGF- β , and IL-35, or by enhanced expression of (ii) ectoenzymes CD39 and CD73 or (iii) TIGIT (Figure 1Ad). CD39 and CD73 mediate a gradual hydrolysis of danger signals of ATP and ADP to anti-inflammatory adenosine, which binds to the A2A receptor of effector cells and induces their inhibition [56]. CD39 can be up-regulated on V δ 1 and V δ 2 TIL after their activation with phorbol ester and ionomycin [31]. TIGIT is often up-regulated in Tregs and marks a highly suppressive Treg phenotype. TIGIT has been reported to inhibit T cells via several mechanisms [57]. The influence of Treg on $\gamma\delta$ TIL is not well defined.

Tumor cells or stromal cells (e.g., MDSC) of the TME over-express indoleamine-2,3-dioxygenase (IDO), which induce an accumulation of IDO-downstream molecule kynurenine. Further arginase production is often enhanced in tumor cells or MDSC. An increased kynurenine or arginase production inhibits the degranulation and cytotoxicity of $\gamma\delta$ T cells and thereby their anti-tumor response [58, 59]. IDO overexpression in tumor cells and other cells of the TME is mediated by the release of PGE2 and IL-10, but also by IFN- γ , recruits Treg, and induces tolerogenic T cell responses (Figure 1Ac) [60].

Additionally, glycan-lectin interactions that include sialic acid-binding immunoglobulin-like lectins (siglecs) and C-type lectin receptors such as selectins or galectins modulate the pro- or anti-tumor function of $\gamma\delta$ T cells. Galectins are produced by $\gamma\delta$ T cells, tumor cells, or tumor-surrounding cells and are often enhanced after crosstalk with each other. Because galectin-1 and galectin-9 have more pro-tumoral effects on $\gamma\delta$ T cells, galectin-3 did not influence their cytotoxicity but their proliferation in a different way [2, 35, 61]. Although V δ 1 T cell proliferation was not modulated by galectin-3, V δ 2 T cell proliferation was inhibited by enhanced galectin-3-producing tumor cells after cross-talk of $\gamma\delta$ T cells and tumor cells (Figure 1Ae; Figure 1Bd) [35, 61]. The different influence of galectin-3 on the V δ 1 versus V δ 2 T cell proliferation can be an advantage for a V δ 1 T cell-based immunotherapy or one reason for the described exhaustion of V δ 2 TIL. V δ 2 T cells within PBL, which expand selectively after in vivo stimulation with aminobisphosphonates (*n*-BP), are suggested to migrate to the tumor site. Comparable to V δ 2 TIL, migrating V δ 2 T cells could be inhibited in their proliferation after cross-talk with tumor cells or tumor-surrounding cells producing high amounts of galectin-3, which is an important aspect for the optimization of $\gamma\delta$ T cell subset-based immunotherapies [61].

5 | $\gamma\delta$ T Cells for Future Immunotherapies

Several clinical trials have been conducted in the past decades to investigate the utilization of $\gamma\delta$ T cells in cancer immunotherapy [62]. For example, V δ 2 T cells could be successfully expanded in vivo after application of synthetic *n*-BPs (e.g., zoledronate) in combination with low doses of interleukin-2 (IL-2) in non-Hodgkin lymphoma/multiple myeloma [63] or prostate cancer [64]. Additionally, autologous transfers of ex vivo expanded $\gamma\delta$ T cells were examined, for example, in renal cell carcinoma [65]. Importantly, no safety issues were identified in such early-phase clinical trials [66], but, unfortunately, they showed limited clinical efficacy, if at all [62]. This might be due to induction of senescent- and/or exhaustion-associated programs in the targeted, mostly V γ 9V δ 2 T cell subsets. Thus, the “first-generation attempts” to exploit $\gamma\delta$ T cells’ unique features successfully in the clinic has been proven very difficult, but “next-generation” studies show promising data in preclinical and early clinical observations.

These are, for example, bsAb, combining a tumor and $\gamma\delta$ TCR binding site and thereby supporting colocalization of effector and target cells in vivo. Oberg et al. described for the first time such a bsAb, a [(HER2)₂ × V γ 9] tribody, as an engager of T cell-mediated cytotoxicity against ductal adenocarcinoma pancreatic cells and ovarian cancer cells or other HER2-expressing tumor entities [7, 8]. Apart from these $\gamma\delta$ TCR-specific bsAbs, further constructs binding non-exclusively expressed receptors on the surface of $\gamma\delta$ T cells, for example, [(HER2)₂ × CD16], [MICA × CD20], or [ULBP × CD20], revealed significant cytotoxicity against tumor cells [67–69]. Similar promising results were conducted by engagers binding to CD40 and V γ 9V δ 2 TCR in vitro and in mice [70], CD1d and V γ 9V δ 2 TCR in vitro [71], or as tribody targeting CD19 with two binding sites and CD16 with one, activating thereby NK and $\gamma\delta$ T cells in vitro [72] in hematologic malignancies.

Due to the relatively low occurrence of $\gamma\delta$ T cells in the peripheral blood, expansions are necessary for other cell therapeutic approaches. V γ 9V δ 2 T cells can be expanded via pyrophosphates or *n*-BP combined with further substances like IL-2, and IL-18 out of peripheral blood mononuclear cells (PBMCs) and do per se not require an isolation upfront. The combination of zoledronate and IL-2 with Vitamin C in a novel expansion protocol resulted in higher proliferative activity and cytotoxic potential of V γ 9V δ 2 T cells against cancer cell lines [73] and indicated in a clinical trial an association with prolonged survival in a few advanced liver and lung cancer patients [74]. However, in vitro expansions targeting the V γ 9V δ 2 TCR harbor the risk of generating immunosuppressive cells [75]. Apart from V γ 9V δ 2 T cells, also V δ 1 T cells have been considered for therapeutic exploitation [3]. A major obstacle here is their very low abundance in the blood and the lack of knowledge of the vast majority of their ligands. The clinical-grade “Delta One T cell (DOT)-protocol,” for example, expands and differentiates V δ 1 T cells into effector cells expressing high levels of NK cell receptors (NKp30, NKG2D, or DNAM-1) and revealed promising results in leukemia mouse models [76–78]. The DNAM-1 ligands PVR and B7-H6 have recently been described as being involved in AML rejection by DOT-cells [79]. On the basis of the latter, a clinical trial in

AML is currently ongoing (NCT05886491). Davies et al. described recently an explant system as an alternative approach to blood-based expansion methods to generate larger numbers of V δ 1 T cells out of skin tissues with a putative potential to better infiltrate solid cancers, although those yielded numbers are not sufficient to generate clinical products [80].

The high cytotoxicity and plasticity of $\gamma\delta$ T cells and their potential to be used in an allogeneic therapy setting aroused interest to also exploit these cells genetically modified for immunotherapies. For example, anti-CD19 V γ 9V δ 2 CARs and later also pooled V δ 1 and V δ 2 anti-CD19 CARs revealed in vitro and in vivo (mouse) anti-cancer reactivities [81]. An anti-NKG2D V γ 9V δ 2 CAR has proven significant reduction of tumor burden in xenograft mouse models based on ovarian and CRC cell lines [82] and is currently investigated in a clinical trial (NCT05302037). Similar results were revealed also for the V δ 2-negative compartment in hematologic malignancies with, for example, transduced DOT cells [78] or anti-CD20 CARs in V δ 1 T cells [83]. The latter is currently investigated also in a clinical trial (NCT04735471).

Alternative approaches focus on the $\gamma\delta$ TCR, for example, as soluble bispecific molecule $\gamma\delta$ TCR anti-CD3 bispecific molecules (GABs) or as $\gamma\delta$ TCR engineered $\alpha\beta$ T cells [84, 85]. For example, V γ 9V δ 2 transduced $\alpha\beta$ T cells target hematologic but also solid tumor cell lines [85] and are currently observed in a clinical trial in multiple myeloma (NCT04688853). Further attempts to exploit $\gamma\delta$ T cells for future cancer treatment approaches have been reviewed recently [3, 86].

6 | Future Perspectives

Human $\gamma\delta$ T cells combine unique anti-tumor features in most cancer entities, in particular the non-V δ 2 T cell subset. V δ 1 T cells are predictors for favorable clinical outcomes [40, 41] and promising candidates for various novel treatment attempts in pre-clinical and clinical investigations. Additional studies are required to better understand the individual role and function of specific $\gamma\delta$ T cell subsets in the respective TME. In particular, the identification of expression patterns of receptors guiding $\gamma\delta$ T cells into healthy and malignant tissues and a better understanding of the groundling mechanisms are of uttermost importance for a successful leverage of these cells in future ICB. In particular, this involves not only the exploring of infiltration patterns but also the spatial vicinities of $\gamma\delta$ T cell subtypes and their direct and indirect cellular interactions to gain a better understanding of when anti- or pro-tumor functionalities are induced/promoted. This is independent and fundamental for any kind of utilization of $\gamma\delta$ T in future immunotherapy strategies.

Acknowledgments

We gratefully thank our teams in Tübingen, Mannheim and Kiel, and our collaboration partners for excellent and inspiring discussions. The author(s) declare financial support was received for the research, authorship, and/or publication of this article. This work was supported by the Forschungsgruppe FOR2799 "Receiving and Translating Signals via the $\gamma\delta$ TCR" from the Deutsche Forschungsgemeinschaft (DFG, WE 3559/6-2 to DW and WI 5021/2-2 to KWH).

Open access funding enabled and organized by Projekt DEAL.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

We did not include "Data Availability Statement" since data sharing is not applicable to this review as no datasets were generated or analyzed during the current study.

Peer Review

The peer review history for this article is available at <https://publons.com/publon/10.1002/eji.202451075>.

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